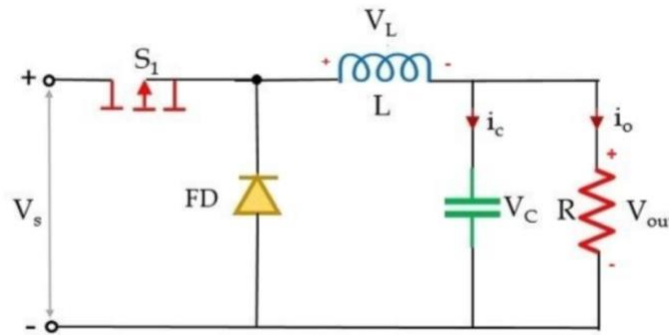


Objective

The objective of this case study is to design a buck converter operating in Continuous Conduction Mode (CCM) with an input voltage range of 12-15 V, producing a regulated output voltage of 8 V, capable of delivering a load power of 1.5 W.

Circuit Diagram



Working

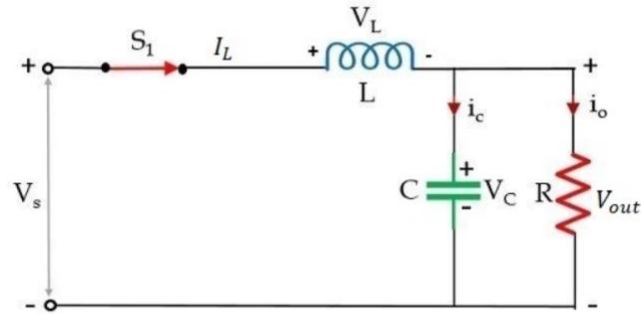
The operation of a buck converter relies on the principle of energy storage in an inductor. The voltage drop across the inductor is directly proportional to the rate of change of current flowing through it. A switching transistor is employed between the input and output, continuously turning on and off at a high frequency. This switching action, combined with the energy stored in the inductor, ensures a steady and regulated output voltage.

The input voltage is applied through a controllable solid-state device functioning as a switch. Additionally, a freewheeling diode is incorporated into the circuit to facilitate the current flow during the transistor's off-state. The inductor and capacitor work together to minimize current and voltage ripples, thereby stabilizing the DC output. In this configuration, the load is purely resistive, and both the input voltage and load current remain constant. Effectively, the load behaves like a current source.

There are two modes of operation of the Buck converter. They are:

1. Mode I: Switch 1 is ON and Diode is OFF
2. Mode II: Switch 1 is OFF and Diode is ON

Mode I: Switch 1 is ON and Diode is OFF



In this mode of operation, switch S_1 is in closed condition i.e. ON state, and diode is in open condition i.e. OFF state. Thus switch S_1 allows the flow of current through it. The circuit diagram for this mode is shown in the above figure.

When a constant dc voltage is applied across the input terminal of the circuit, then the current flow in the circuit through closed switch S_1 whereas the diode is in reverse biased condition. Due to this current flow in the circuit, the inductor “L” store energy in the form of a magnetic field. The capacitor is connected as shown in the circuit diagram. The current flows through the capacitor also and hence the capacitor store the charge. The voltage across the capacitor appears across the load and is equal to the output voltage V_o .

Let us say the switch S_1 is ON for a time T_{on} and it is OFF for a time T_{off} .

Then the total time period T is a combination of T_{on} and T_{off} time.

$$\text{i.e. } T = T_{on} + T_{off}$$

The duty cycle is given by:

$$D = \frac{T_{on}}{T}$$

Applying KVL in the above circuit, we get:

$$V_s = V_L + V_{out}$$

$$V_L = V_s - V_{out}$$

$$V_L = L \frac{dI_L}{dt}$$

$$V_L = L \frac{dI_L}{dt} = V_s - V_{out}$$

$$\frac{dI_L}{dt} = \frac{V_s - V_{out}}{L}$$

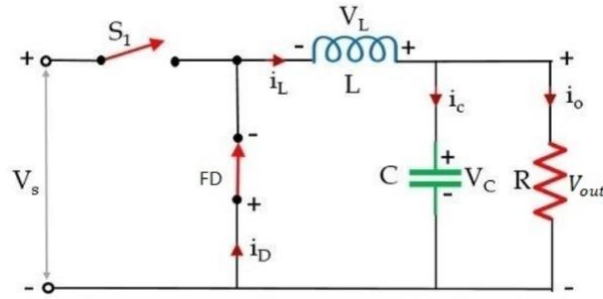
$$\frac{\Delta I_L}{\Delta t} = \frac{V_s - V_{out}}{L}$$

$$\frac{\Delta I_L}{DT} = \frac{V_s - V_{out}}{L}$$

$$\Delta I_L = \left(\frac{V_s - V_{out}}{L} \right) DT$$

This equation gives the change in current through the circuit when switch S1 is closed.

Mode II: Switch 1 is OFF and Diode is ON



In this mode of operation, the switch S1 is in open condition i.e. OFF state and diode is in closed condition i.e. ON state. Thus switch diode allows the flow of current through it, whereas switch S1 blocks the current flow through it. The circuit diagram for this mode is shown in the above figure.

We know that the inductor in the circuit store energy in the form of a magnetic field. When S1 is open, then the inductor acts as the source. Hence diode becomes closed. In this mode of operation, the inductor releases the energy stored in the previous mode when switch S1 was closed. During releasing of energy stored in the inductor, the polarity of the inductor gets reversed which causes the freewheeling diode to come in forward biased condition. So it allows the flow of current in the circuit through diode.

The flow of current in the circuit will continue until the stored energy in the inductor gets completely collapsed. When the inductor gets completely discharged, the freewheeling diode comes in reverse biased condition and hence diode will be open. Instantly, the switch S1 will get closed. In this way, the cycle will be continuing.

Now, applying KVL in the above circuit,

$$V_L + V_{out} = 0$$

$$V_L = -V_{out}$$

$$V_L = L \frac{di_L}{dt}$$

$$V_L = L \frac{di_L}{dt} = -V_{out}$$

We know

$$T = T_{on} + T_{off}$$

Thus

$$T = DT + T_{off}$$

$$T - DT = T_{off}$$

$$(1-D)T = T_{off}$$

$$T_{off} = (1-D)T$$

$$T_{off} = \Delta t = (1-D)T$$

$$V_L = L \frac{di_L}{dt} = -V_{out}$$

$$L \frac{\Delta i_L}{(1-D)T} = -V_{out}$$

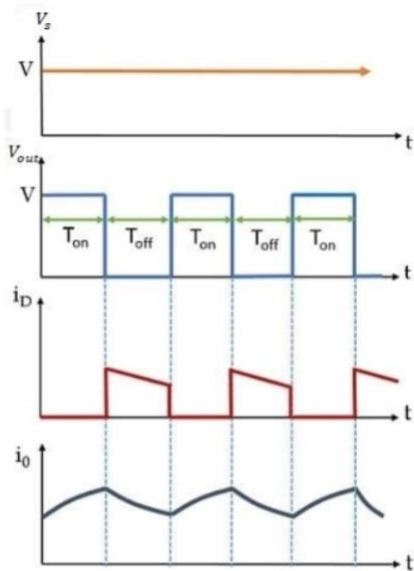
$$\Delta i_L = -\frac{V_{out}}{L}(1-D)T$$

Since the net change in current through the inductor in one complete cycle is zero i.e, the summation of the rate of change of current in Mode I and Mode II becomes zero.

$$\left[\left(\frac{V_s - V_{out}}{L} \right) DT \right] + \left[-\frac{V_{out}}{L}(1-D)T \right] = 0$$

$$\frac{V_{out}}{V_s} = D$$

Waveforms:



Application

Buck converter is basically used for conversion of high dc voltage to low dc voltage. It is mostly used in the following applications:

- Solar charging systems for efficient energy conversion.
- Battery-powered systems such as chargers for mobile phones, laptops, and power banks.
- Power audio amplifiers for stable and efficient power delivery.
- Point-of-load converters in PCs and motherboards.
- Self-regulating power supplies, like Switch-Mode Power Supplies (SMPS).
- Adaptive control systems requiring precise voltage regulation.

Design

Given:

$$V_{in} = 12\text{ V}$$

$$V_o = 8\text{ V}$$

$$P_o = 1.5\text{ W}$$

Assuming

$$f = 10\text{ kHz}$$

$$\frac{\Delta V_o}{V_o} = 0.01$$

Duty Cycle:

$$D = \frac{V_o}{V_{in}} = \frac{8}{12} = 0.66$$

Resistance:

$$R = \frac{V_o^2}{P_o} = \frac{64}{1.5} = 42.66 \text{ ohm}$$

Approximating the resistor value to 43 ohms.

Inductance:

$$L_{cal} = \frac{(1-D)R}{2f} = \frac{(1-0.666) * 42.66}{2 * 10 * 10^3} = 7.11 * 10^{-4} H$$

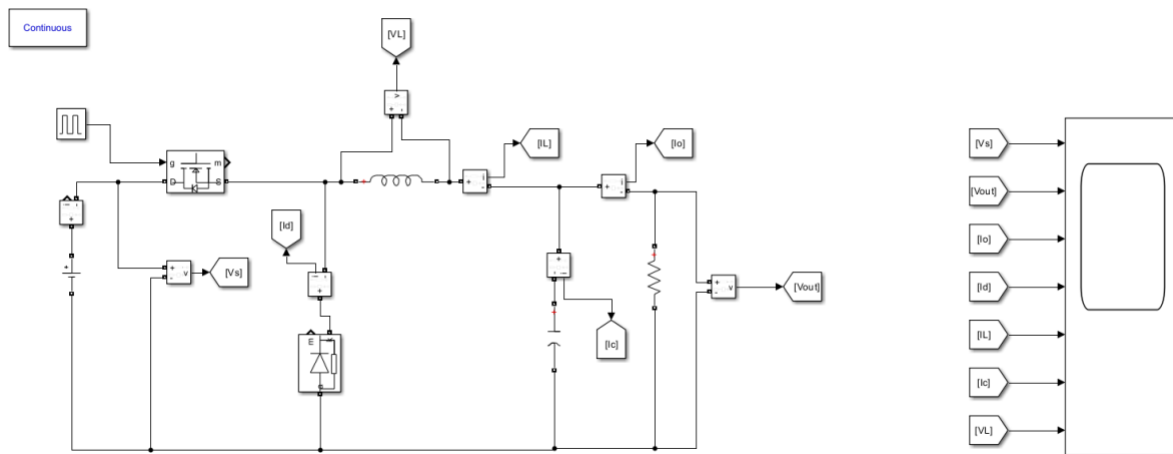
$$L_{design} = 10 * L_{cal} = 7.11 mH$$

Capacitance:

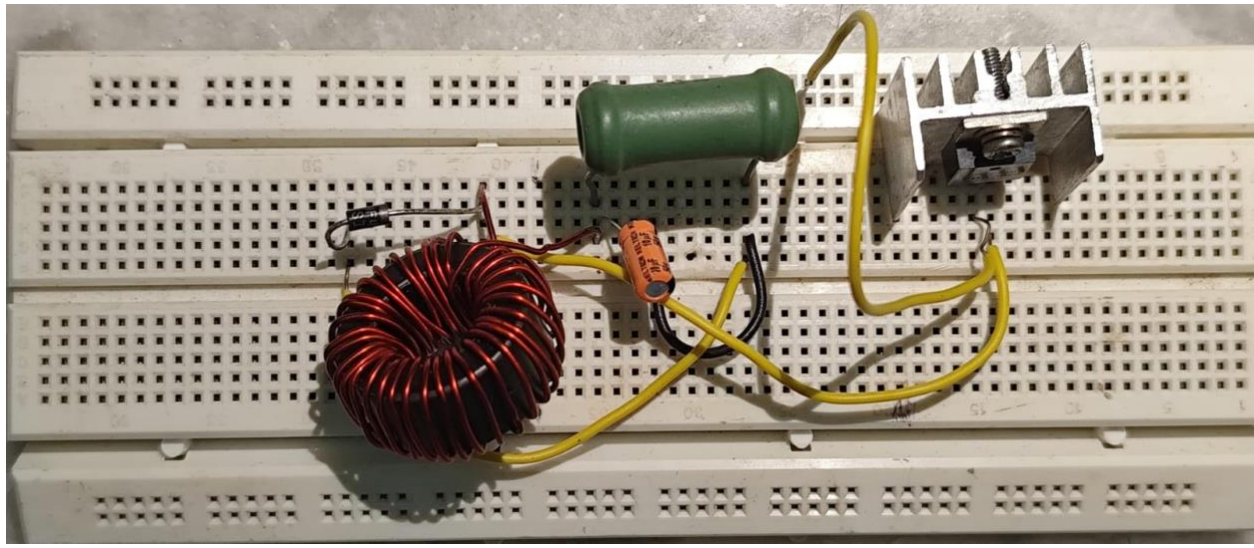
$$C_{cal} = \frac{1-D}{8 * L * \frac{\Delta V_o}{V_o} * f^2} = \frac{1-0.666}{8 * 7.11 * 10^{-3} * 0.01 * 10^8} = 0.058 mF$$

$$C_{design} = 10 * C_{cal} = 0.58 mF$$

Simulation figure



Hardware Figure



Result

The designed buck converter operates in Continuous Conduction Mode (CCM), stepping down an input voltage range of 12-15 V to a stable output of 8 V. With a load power of 1.5 W, the converter ensures efficient voltage regulation and is suitable for applications requiring consistent and reliable output under varying input conditions. Operating in CCM also enhances output stability by reducing voltage and current ripples.